



Department of Mathematics and Natural Sciences

Semester: Fall 2023

Course Title: Mathematics II: Integral Calculus and Differential Equations

Course Code: MAT- 120

Total marks: 20

Time: 1 hour

Instructions:

- Please write your name, ID, and section on your script
- Give page number (use pen) in the right corner of each page
- Please read the question paper very carefully
- Answer all the questions

1. Use Riemann sum to find the area between the curve $f(x) = x^2 - 2x + 3$ and the interval $[1, 3]$. Use x_k^* as the right endpoint of each subinterval. [4]
2. Determine whether the given improper integral $\int_{-1}^2 \frac{x}{(x+1)^2} dx$ is convergent or divergent. [4]
3. Use Trigonometric substitution to evaluate $\int_{1/4}^{\sqrt{3}/4} \sqrt{1-4x^2} dx$. [4]
4. Use Gamma, Beta functions to evaluate $\int_0^1 \frac{x^3}{\sqrt{1-x^3}} dx$. [4]
5. Find the area of the surface that is generated by revolving the curve $x = \sqrt[3]{y}$ between $y = 1$ and $y = 8$ about x- axis. [4]